

Rashmi Gunawardana

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EDUCATION

Bachelor of Science in Information Systems with Honours

University of Colombo School of Computing

May 2021 – May 2025

Colombo, Sri Lanka

- **CGPA : 3.7/4**
- Achievements : Director's List 2022/2023

WORK EXPERIENCE

Undergraduate Teaching Assistant

University of Colombo School of Computing

June 2024 – Present

Colombo, Sri Lanka

- Taught the subject "SCS1303-Introduction to Software Engineering" to first-year students following Computer Science Major.
- Teaching the subject "SCS2211: Laboratory II" to second-year students following Computer Science Major.
- Helps the lecturer and assistant lecturers to conduct lecture sessions and prepare lecture materials.

Software Engineer Intern

Millennium IT ESP

Nov. 2023 – May 2024

Colombo, Sri Lanka

- Worked on project Ayati, a system made for a local hospital which specialized in treating children with disabilities. I have conducted research and designed high level architecture diagrams for the project.
- Implemented Ayati web application with java back-end, javascript front-end and postgresql hosted database.
- Additionally I Worked as a developer for the company's own front-end framework, developed components and maintained libraries enhancing its functionality and usability.

Blogger

Medium

Mar. 2021 – Present

- Have done technical writing for different platforms and organizations.
- The Blog has grown to **100+ followers** in the medium platform.
- Mainly writes on Microsoft azure platform, Azure Machine Learning studio and CI/CD technologies.

RESEARCH EXPERIENCE

Enhancing Personalized Learning Interfaces with Generative AI for Individuals with ADHD

In Association with University of Colombo School of Computing

August 2024 – Present

- This research, supervised by Dr. R.A.C. Ranasinghe, aims to develop an innovative learning platform tailored for individuals with ADHD. By applying principles of Human-Computer Interaction (HCI), we compare three distinct learning management systems (LMS): a conventional LMS used by the general population, a manually optimized LMS designed specifically for ADHD users, and an AI-generated LMS customized to ADHD needs through enhanced prompting techniques. This study involves 20 ADHD students whose feedback on each LMS informs our analysis, allowing us to assess how AI-driven customization can adapt interfaces to diverse cognitive and learning styles. By integrating stable diffusion-infused techniques and low-code/no-code UI generation, this project pushes the boundaries of user-centric design in educational technology, offering a more inclusive and effective learning experience for individuals with ADHD.

Secure and Seamless: A Literature Review of Passwordless Authentication Techniques and Implementation

Associated with University of Colombo School of Computing

- Collaborated on a comprehensive literature review exploring passwordless authentication methods, including biometric systems, token-based authentication, and behavioral analytics, under the supervision of Prof. Kasun de Zoysa. This literature review was a requirement to pass module IS3110 (Research Methods) and achieved a **Grade A** result.
- Analyzed the effectiveness and implementation challenges of emerging authentication technologies, focusing on their potential to enhance security and user experience.
- Identified best practices and key trends in the field, contributing valuable insights to the academic discourse on modern authentication strategies and cybersecurity.

TECHNICAL PROJECTS

Earthquake Data Engineering Project with Microsoft Fabric

- Built an end-to-end data engineering and analytics pipeline for processing and analyzing earthquake event data from USGS. Utilized Microsoft Fabric components, including Data Factory, Data Engineering, and Power BI, to ingest, transform, and visualize data. Leveraged Python and PySpark for data transformation and ensured comprehensive insights through interactive Power BI dashboards.

Deploying Machine Learning Models with Azure Machine Learning Studio

- Designed and implemented machine learning workflows for data preprocessing, model training, and evaluation using Python and Jupyter Notebooks. Utilized Azure Machine Learning Studio to streamline the deployment process, ensuring robust, scalable, and reproducible model performance. This project highlighted the integration of cloud-based tools for AI-driven data solutions.

OpinionLk - A Survey Management Platform

- This is an online survey-filling platform developed using MongoDB, Express, React, and Node technologies. Azure Hosting services were used to host this project and Azure CosmosDB for MongoDB for the database. The back-end was hosted using the Heroku hosting platform.

Mom Care - A Platform For Managing the Pregnancy Journey of Women

- This is a platform to handle the pregnancy journey of mothers to be, that handles all the pre-birth activities that need to be done with Midwives and nurses at the hospital. developed using HTML , CSS , JavaScript, Java and Python.

CERTIFICATIONS

AI -102 : Microsoft Certified: Azure AI Engineer Associate

AI-900 : Microsoft Certified: Azure AI Fundamentals

AZ-900 : Microsoft Certified: Azure Fundamentals

MB-910: Microsoft Dynamics 365 Fundamentals (CRM)

ACA System Operator Certification by Alibaba Cloud Academy

Postman API Fundamentals Student Expert Certification

TECHNICAL SKILLS

Programming Languages : Python , Java , JavaScript , ReactJs , HTML , CSS , SQL , MySQL

Databases : MongoDB , Azure Cosmos DB , PostgreSQL

Other Familiar Platforms and Tools : Jupiter Notebooks , Azure AI Studio , Azure Machine Learning Studio , Microsoft Fabric , Azure DevOps , Google Colab , Hugging face

ACHIVEMENTS

- Finalist in the I am a Researcher Competition at the 24th ICTer International Conference.
- Most Outstanding Volunteer, Sub-Committees 2022/2023, IEEE Young Professionals Sri Lanka.
- Participant in Microsoft Code Without Barriers 2023, organized by iTrain Asia with Microsoft.
- Finalist in Hackholics Hackathon, organized by the WIE Student Branch of UCSC.
- Participant in IEEEExtreme 16.0, organized by IEEE.
- 1st Runner-Up in Emerging Entrepreneur 2.0 (Tourism Industry), organized by AIESEC in the University of Colombo.
- 2nd Runner-Up in Parallax 2021 UI/UX Hackathon, organized by the Rotaract Club of IIT.
- Finalist in Madhack 2021 Mobile Development Hackathon, organized by IEEE Student Branch in UCSC.
- 2nd Place in Mechtrogon Ideathon 2021, organized by AIESEC in the University of Colombo.

VOLUNTEERING EXPERIENCE

Gold Level Microsoft Learn Student Ambassador

- First Female Gold Ambassador in Sri Lanka. As a Gold Level Microsoft Learn Student Ambassador, I have actively contributed to advancing AI education by planning and conducting over **20 sessions** for more than **1000 students** across various platforms, including Microsoft Sri Lanka and my university. My sessions have covered a range of cutting-edge topics such as GitHub Actions, MLOps, deploying Machine Learning models using Azure Machine Learning Studio, and AI-driven data analytics with Microsoft Fabric.
- In addition to educational outreach, I have hands-on experience with Microsoft Generative AI tools and data technologies, including designing, developing, and maintaining data pipelines using Azure Data Factory and generating insights through Power BI. I specialize in optimizing ETL processes to enhance data extraction, transformation, and loading efficiency, further showcasing my proficiency in Microsofts ecosystem for AI and data-driven solutions.

Vice-Chair at IEEE Techverse Sri Lanka

- As Vice-Chair of IEEE Techverse Sri Lanka, I demonstrated strong leadership by managing and coordinating a diverse team of **32 members**. My role involved strategic planning, delegation of tasks, and fostering collaboration within the team to successfully organize **10+ workshops, webinars, and various events** under the IEEE Sri Lanka Section.
- I played a key role in driving engagement and organizing events at IEEE Techverse Sri Lanka, focusing on technology trends and professional development. My leadership boosted the chapters visibility, expanded skill development opportunities, and involved effective project management and team-building skills in high-pressure environments.

REFERENCES

Name: Dr. R.A.C Ranasinghe

Senior Lecturer, University of Colombo School of Computing

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Name: Dr. K.A.K.T Karunanayaka

Senior Lecturer, Department of Communication and Media Technologies

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